Hebbal Edge PE Hardware/Environmental Fault Resolution SOP (Ericsson CLI Dialect)

Node: HEB-PE-12 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_9179

Description

The Hebbal Edge PE hardware/environmental fault, characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss, occurs due to the inherent complexities of the Ericsson PE's hardware architecture. Specifically, the PE's reliance on ASICs (Application-Specific Integrated Circuits) for high-speed packet processing and forwarding results in a heightened susceptibility to memory parity errors. Furthermore, the PE's modular design, which includes line cards for increased scalability and flexibility, can lead to line card crashes if not properly maintained or upgraded. Additionally, the PE's power supply redundancy mechanism, designed to ensure high availability, can be compromised by environmental factors such as power fluctuations or overheating, resulting in a loss of redundancy. This SOP aims to provide a step-by-step guide for resolving these hardware/environmental faults on the Ericsson PE at the Hebbal Edge site, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Verify that the site is physically accessible and that all necessary personnel have been dispatched to the Hebbal Edge site.

- Check the redundant link status to ensure that it is functioning correctly and can handle the increased traffic load.

- Conduct a thorough inspection of the PE equipment, including the line cards and power supplies, to identify any visible signs of damage or malfunction.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware failure or environmental stress	High
Line card crashes due to software or hardware issues	Medium
Power supply redundancy loss due to environmental factors or equipment failure	Low

Diagnostic Commands & Expected Outputs

```
Command: show controllers asic memory, Expected Output: Memory Status: OK, Parity Errors: 0, Description: This command displays the current status of the ASIC memory, including any parity errors. A parity error indicates a hardware failure or environmental stress that requires immediate attention.
```

```
Command: show card status, Expected Output: Card Status: OK, Line Card: [insert line card number], Description: This command displays the current status of the line cards, including any issues or faults. A line card crash can cause packet loss and network downtime, necessitating prompt resolution.

Command: show power supply status, Expected Output: Power Supply Status: OK, Redundancy: [insert redundancy status], Description: This command displays the current status of the power supplies, including the redundancy status. A loss of redundancy can compromise network availability, requiring immediate attention.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC or line card, Command: install card [insert card number], Description: This command replaces the faulty ASIC or line card with a new one, resolving the hardware failure or environmental stress.

Step 2: Step: Reset power supply redundancy, Command: reset power supply redundancy, Description: This command resets the power supply redundancy mechanism, ensuring that the network can operate in redundancy mode and preventing loss of availability.

Post-Check Verification

```
Command: show controllers asic memory, Expected Output: Memory Status: OK, Parity Errors: 0, Description: This command verifies that the ASIC memory is functioning correctly and that there are no parity errors.

Command: show card status, Expected Output: Card Status: OK, Line Card: [insert line card number], Description: This command verifies that the line card is functioning correctly and that there are no issues or faults.

Command: show power supply status, Expected Output: Power Supply Status: OK, Redundancy: [insert redundancy status], Description: This command verifies that the power supplies are functioning correctly and that the redundancy status is as expected.
```

Roll-back Procedures

- Step: Restore original configuration, Command: configure replace [insert configuration file], Description: This command restores the original configuration, ensuring that the network returns to its previous state in case the fix fails.

- Step: Revert changes, Command: rollback [insert rollback command], Description: This command reverts the changes made during the fix, ensuring that the network returns to its previous state in case the fix fails.